Level 2 Functional Skills Mathematics
SAMPLE PAPER 2

City & Guilds
A City & Guilds Group Business
Version 1.1

Duration: 25 minutes
Total marks: 15 marks

MODEL SOLUTION SCORE
15 / 15 marks (100% Pass)
25 minutes · Verified by MFS

SECTION 1 – CALCULATOR NOT PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

Please check that your name is correctly printed on the candidate barcode label.
If not, please tell the invigilator before the start of the exam.

You should have the following for this assessment:

- a pen with black or blue ink
- a pencil (for diagrams, graphs and charts only)
- an eraser
- a 30cm ruler.



You must NOT use a protractor.
You must NOT use a calculator for Section 1.

General instructions

- Read through each question carefully.
- Write all your answers in this booklet.
- Check your calculations and check that your answers make sense.

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SECTION 1 – CALCULATOR NOT PERMITTED

There are **15** marks available in this section.

You should check all your work as you go along.

You must **not** use a calculator in this section.



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Q1

What is 108 as a fraction of 648? Give your answer in its simplest form.

Method: Write as fraction and simplify by HCF (108):

108 / 648 = (108 ÷ 36) / (648 ÷ 36) = 3 / 18

3 / 18 = (3 ÷ 3) / (18 ÷ 3) = 1 / 6

$\frac{1}{6}$ [B1]

(1 mark)

Q2

Which one of the following lists is in decreasing order?

(tick one box)

A $\frac{1}{5}$ 0.05 $\frac{17}{50}$ 0.15 ☐

B 0.05 $\frac{17}{50}$ $\frac{1}{5}$ 0.15 ☐

C 0.15 $\frac{1}{5}$ 0.05 $\frac{17}{50}$ ☐

D $\frac{17}{50}$ $\frac{1}{5}$ 0.15 0.05 ☑ [B1]

Convert all values to decimals to compare:

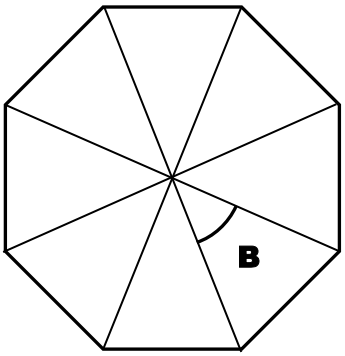
17/50 = 0.34 · 1/5 = 0.20 · 0.15 · 0.05

Decreasing: 0.34 > 0.20 > 0.15 > 0.05 --> Option D

(1 mark)

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Q3 The diagram shows a regular polygon.



What is the size of angle B.

[B1]

Regular octagon has 8 exterior angles adding to 360°:

Exterior angle B = 360° ÷ 8 = 45°

45 °

(1 mark)

Q4

Work out the value of y if 4y = 144

[B1]

Divide both sides by 4:

y = 144 ÷ 4 = 36

36

(1 mark)

Q5

$\frac{3^2}{3} - 28 =$

[B1]

BIDMAS: Index / power first, then division, then subtraction:

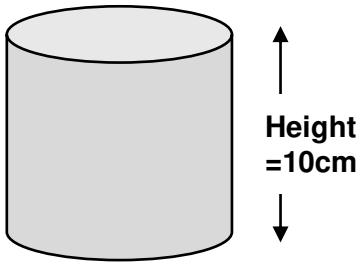
$3^2 \div 3 - 28 = 9 \div 3 - 28 = 3 - 28 = -25$

-25

(1 mark)

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Q6 The radius of the top of this cylinder is **7cm**



Use

$$\pi = \frac{22}{7}$$

What is the volume of the cylinder?

Formula: Volume = (22/7) × r^2 × h (where pi = 22/7):

V = (22/7) × 7² × 10 = (22/7) × 49 × 10 = 22 × 7 × 10 = 1540 cm³

[BI]

1540 _____ cm^3

(1 mark)

Q7

$\frac{11}{8} - \frac{1}{16} =$

(tick one box)

A $\frac{5}{8}$ ☐

B $1\frac{5}{8}$ ☐

C $1\frac{5}{16}$ ☑

D $2\frac{5}{16}$ ☐

Common denominator (LCM = 16):

11/8 = 22/16

22/16 - 1/16 = 21/16 = 1 5/16

[BI]

(1 mark)

Q8

4.50 ÷ 0.05 =

[BI]

Multiply numerator and denominator by 100:

4.50 ÷ 0.05 = 450 ÷ 5 = 90

90 _____

(1 mark)

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Q9

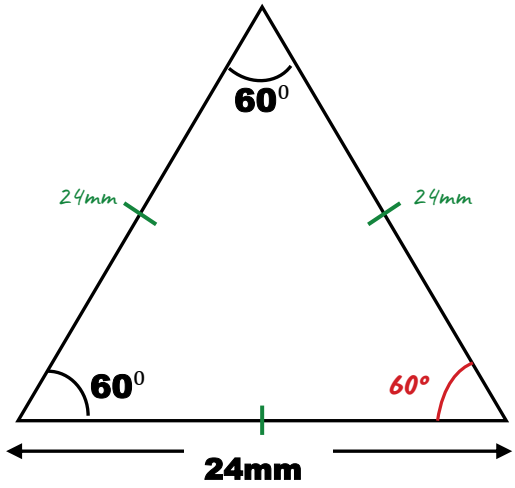


Diagram not to scale

What is the perimeter of this triangle?

Angles in a triangle add to 180°: 180° - 60° - 60° = 60°

All angles are 60° --> Equilateral triangle (all 3 sides equal = 24mm)

Perimeter = 24 + 24 + 24 = 3 x 24 = 72 mm

[BI]

72 _____ mm

(1 mark)

Q10 A call centre aims to deal with calls in less than 5 minutes.

Calls come in randomly.

The table shows data for the calls made to the centre.

Type of call	Proportion of all calls	Completed in less than 5 minutes
Customer complaints	1/4	1/2
New business	3/4	1/8

Work out the probability that the next call will be a customer complaint completed within under 5 minutes.
Give your answer as a fraction in its simplest form.

P(Complaint AND < 5 mins) = P(Complaint) x P(< 5 mins)

= 1/4 x 1/2 = 1 / 8

[BI]

$\frac{1}{8}$

(1 mark)

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Q11 The government announces that the minimum wage for people over 25 years old will increase from £8.21 to £8.72 per hour.

A 26 year old woman works 30 hours a week and is paid the minimum wage.

She thinks that the extra money she will earn will cover a rent increase of £20 per week.

Is the woman correct? Show your calculation.

Decision *(tick one)* **yes** ☐ **no** ☑

Calculation

Hourly wage increase = £8.72 − £8.21 = £0.51 [A1]

Extra earnings per week = 30 hours × £0.51 = £15.30

Comparison: £15.30 is less than £20.00 rent increase (short by £4.70)

Conclusion: The woman is NOT correct.

(1 mark)

Q12 A driver sees this speed limit sign in France. The speed is in kilometres per hour.



He is driving at 80 miles per hour.

He thinks this is below the speed limit.

1 kilometre $= \frac{5}{8}$ mile

Is he correct? Explain your answer showing your calculation.

Decision *(tick one box)* **yes** ☑ **no** ☐

Explanation

Convert 80 mph to kilometres per hour (using 1 km = 5/8 mile): [M1]

80 ÷ (5/8) = 80 × 8 ÷ 5 = 640 ÷ 5 = 128 km/h [A1]

Alternative: 130 km/h × 5/8 = 650 ÷ 8 = 81.25 mph

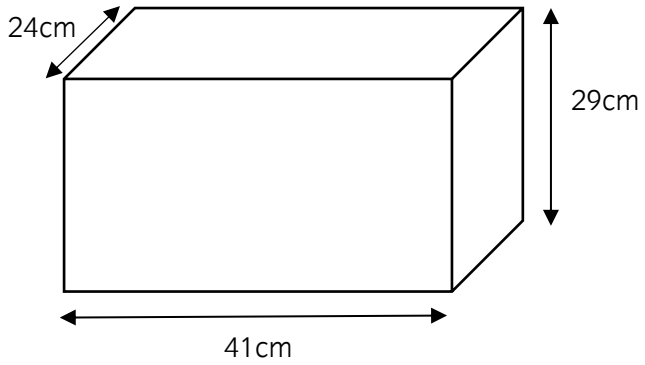
Comparison: 128 km/h < 130 km/h (or 80 mph < 81.25 mph)

The driver is below the speed limit, so he IS correct (Yes).

(2 marks)

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Q13 A man has a fish tank with the following dimensions:



He needs to know approximate volume of the tank.

[M1]

What is its approximate volume?

[A1]

Round dimensions to sensible values (nearest 5 or 10):

24 cm ≈ 25 cm • 29 cm ≈ 30 cm • 41 cm ≈ 40 cm

Approximate Volume = 25 × 30 × 40 = 30,000 cm³

(Acceptable alternative: 20 × 30 × 40 = 24,000 cm³)

30,000 cm³

(2 marks)

End of Section 1

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Duration: 1 hour 20 minutes
Total marks: 45 marks

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins • Verified by MFS

SECTION 2 – CALCULATOR PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

• If you have used any additional answer sheets write the number of additional sheets in this box. ☐
• Please ensure that you **staple** additional answer sheets to the **back** of this booklet, clearly labelling them with your full name, enrolment number, centre number and date in BLOCK CAPITALS.
• You must use a black or blue pen. You may use a pencil for charts and diagrams.
***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

Please check that your name is correctly printed on the candidate barcode label. If not, please tell the invigilator before the start of the exam.

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- a pen with black or blue ink
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- an eraser
- a 30cm ruler.



You must NOT use a protractor.

General instructions

- Read through each question carefully.
- Show your working out (where required).
- Write all your working out and answers in this booklet.
- Check your calculations and check that your answers make sense.
- There are additional pages **including graph paper** at the back of this booklet if you run out of space or ask the invigilator if you need additional sheets of paper.

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SECTION 2 – CALCULATOR PERMITTED

There are **45** marks in this section.

You should check all your work as you go along.

You may use a calculator.



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Q1

0.06 million ÷ 6.2 thousand =

Give your answer to two decimal places

[B1]

0.06 million = 60,000 • 6.2 thousand = 6,200

60,000 ÷ 6,200 = 600 ÷ 62 = 9.677419... --> 9.68 (to 2 d.p.)

9.68

(1 mark)

Q2

What is $\frac{3}{87}$ as a decimal correct to **three decimal places**?

[B1]

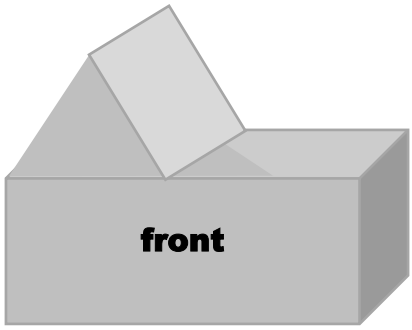
3 ÷ 87 = 0.0344827...

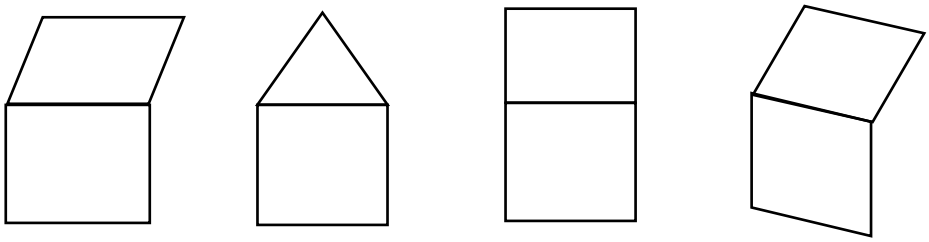
Round to 3 d.p.: 4th digit is 4 (< 5), round down --> 0.034

0.034

(1 mark)

Q3 The diagram shows the outline of a building.





A **B** **C** **D**

Which one of the above shows the elevation of the left side of the building?

(tick one box)

A ☐

B ☐

C ☑

D ☐

Left side view shows the side wall with single sloped roof:

Matches Diagram C perfectly.

[B1]

(1 mark)

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Q4

1m^3 is the same as

(*tick one box*)

A	100 cm^3	☐
B	1000 cm^3	☐
C	100 000 cm^3	☐
D	1 000 000 cm^3	☑

1 m = 100 cm

1 m³ = (100 cm)³ = 100 × 100 × 100 = 1,000,000 cm³

Option D is correct.

[B1]

(1 mark)

Q5 A social club sells 50 pink raffle tickets and 75 yellow raffle tickets.

Tickets are drawn randomly.

What is the probability that the first ticket drawn will be a pink ticket?

Give your answer as a decimal.

Total tickets = 50 + 75 = 125

P(Pink) = 50 / 125 = 2 / 5 = 0.4

[B1]

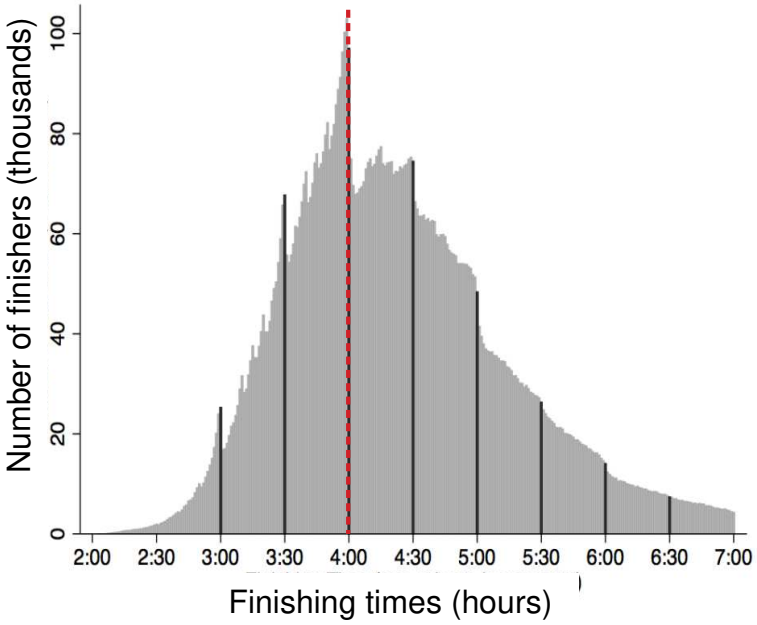
0.4

(1 mark)

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Q6 The chart shows finishing times of marathon runners.

< 4 hours (~10,000) ≥ 4 hours (~30,000)



A report states that most of the runners finished in under 4 hours.

Is the report correct?

Explain your decision.

Decision *(tick one box)* **Yes** ☐ **No** ☑

Explanation

[A1]

Data shows distribution is skewed to the right (longer times):

- Under 4 hours: ~2k (2–3h) + ~8k (3–4h) ≈ 10,000 runners
- 4 hours or more: ~14k (4–5h) + ~12k (5–6h) + ~4k ≈ 30,000 runners

More than half (approx 75%) finished in 4 hours or more.

Therefore, the report is NOT correct.

(1 mark)

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Q7 A shop has a sale.
There is 30% off all black tag items.



One day this notice appears in the shop.



A customer complains that the notice is misleading because it is not true.

Is the customer correct?

Explain your decision. Include calculations to support your decision.

Decision *(tick one box)* **Yes** ☑ No ☐

Show all your working

Step 1: Find original price before 30% reduction: [M1]

£21 represents 70% of original: £21 ÷ 0.70 = £30.00

Step 2: Half of original price = £30.00 ÷ 2 = £15.00

Explanation and supporting calculations.

Step 3: Calculate new promotional price with extra 25% off £21: [M1]

Extra 25% discount = 0.25 × £21 = £5.25

New price = £21.00 − £5.25 = £15.75 (or £21 × 0.75 = £15.75)

Step 4: Compare with half price: £15.75 > £15.00 (75p more expensive!) [A1]

The notice claims "better than half price" which is FALSE. Customer is CORRECT.

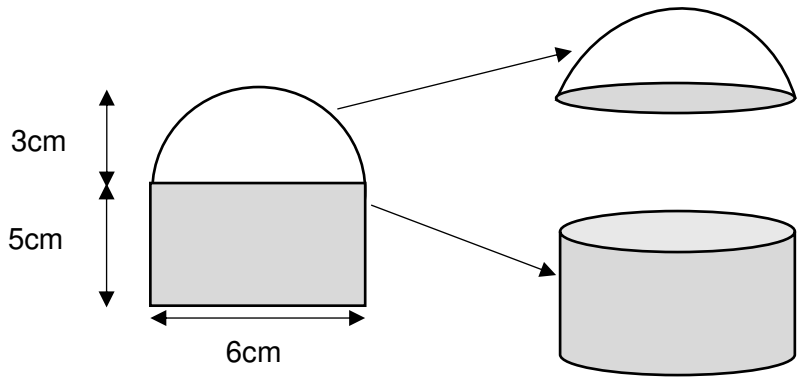
(3 marks)

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Q8 A craftsman uses resin to make a paperweight.

He makes the paper weight from two parts, a hemisphere **and** a cylinder.

He uses this plan.



$$V = \frac{2}{3}\pi r^3$$

V = volume of **hemisphere** in cm^3

r = radius of **hemisphere** in cm

π = 3.14

How much resin does he need for **both** parts?

Show all your working

Step 1: Volume of hemisphere (r = 3 cm, π = 3.14):

V_hemi = (2/3) × π × r³ = (2/3) × 3.14 × 3³ [M1]

V_hemi = (2/3) × 3.14 × 27 = 2 × 3.14 × 9 = 56.52 cm³ [A1]

Step 2: Volume of cylinder (r = 3 cm, h = 5 cm):

V_cyl = π × r² × h = 3.14 × 3² × 5 = 3.14 × 9 × 5 = 141.30 cm³ [M1]

Step 3: Total volume of resin for both parts:

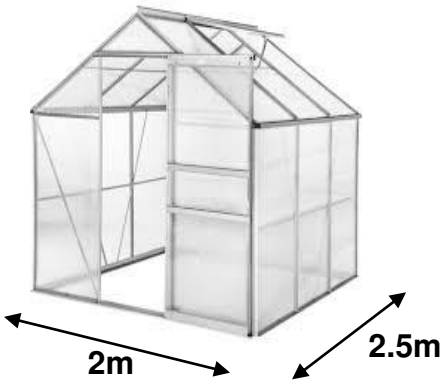
Total = V_hemi + V_cyl = 56.52 + 141.30 = 197.82 cm³ [A1]

Amount of resin 197.82 **cm^3**

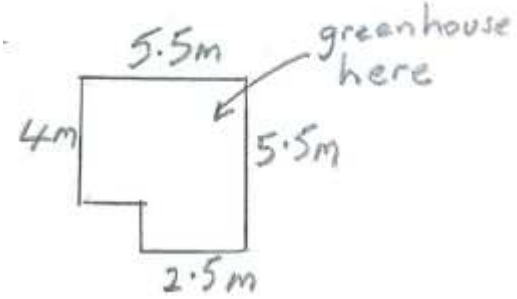
(4 marks)

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Q9 A gardener wants to build this greenhouse in the top right hand corner of her garden.



This is a sketch plan of her garden.



She will leave a 50cm space between the greenhouse and the edge of the garden.

She wants a scaled plan of the garden showing the position of the greenhouse.

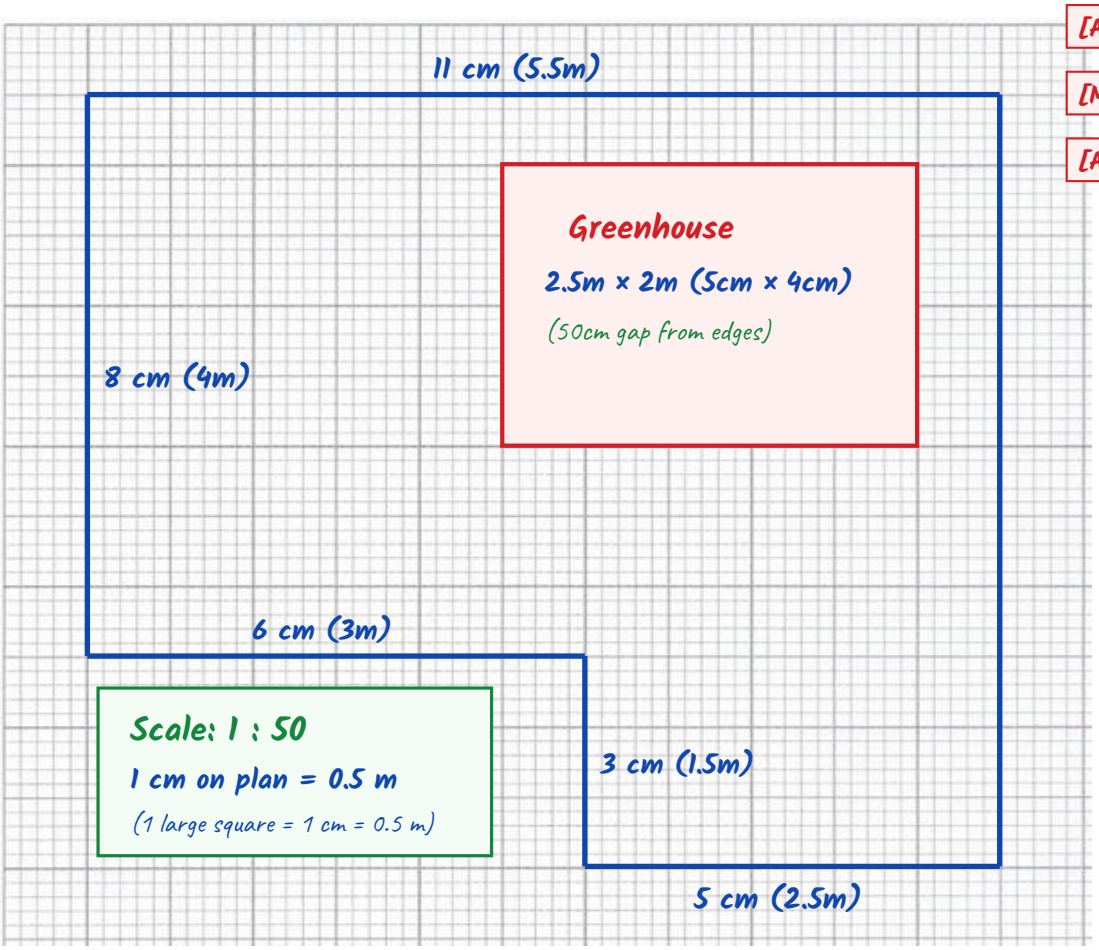
Draw a scale plan. Put the scale you use on the plan.

[M1]

[A1]

[M1]

[A1]



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(4 marks)

There is spare graph paper on page 23

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Q10 A student wants to raise £100 for charity.

He will sell hot dogs at a disco.

He will make 150 hot dogs.

He buys ingredients at *The Supermarket*

The Supermarket	
Hot dog sausages tin of 8	50p
Hot dog buns packet of 6	90p



If he sells all the hot dogs, what is the minimum price he must charge to make a profit of £100?

Show all your working

Step 1: Calculate sausage packs required (tins of 8):

150 ÷ 8 = 18.75 --> Need 19 tins (must round up) [M1]

Cost of sausages = 19 × £0.50 = £9.50

Step 2: Calculate bun packets required (packets of 6):

150 ÷ 6 = 25 packets (exact) [M1]

Cost of buns = 25 × £0.90 = £22.50

Step 3: Total cost of ingredients = £9.50 + £22.50 = £32.00 [M1]

Step 4: Total revenue needed = Costs + Profit = £32.00 + £100.00 = £132.0 [A1]

Step 5: Minimum price per hot dog = £132.00 ÷ 150 = £0.88 (88p)

Minimum price per hot dog _____

£0.88 (88p)

(4 marks)

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Q11 A supermarket buyer compares large eggs from two suppliers.

She wants to buy the largest eggs.

The table shows the masses of a sample of large eggs from Supplier A.

Weights of large eggs (Supplier A)	
Weight in grams	**Number of eggs**
$63 < g \leq 65$	22
$65 < g \leq 67$	27
$67 < g \leq 69$	26
$69 < g \leq 71$	15
$71 < g \leq 73$	10

Midpoint × f = fx

64 × 22 = 1408

66 × 27 = 1782

68 × 26 = 1768

70 × 15 = 1050

72 × 10 = 720

A similar sample taken from supplier B gives a mean value of 66.5g

Which supplier should the buyer use?
Explain your decision.

Decision *(tick one box)* **Supplier A** ☑ **Supplier B** ☐

Show all your working

Total eggs (n) = 22 + 27 + 26 + 15 + 10 = 100 [M1]

Total weight (Σfx) = 1408 + 1782 + 1768 + 1050 + 720 = 6728 g [M1]

Estimated Mean for Supplier A = 6728 ÷ 100 = 67.28 g [A1]

Explanation and supporting calculations.

Comparison: Supplier A mean (67.28 g) > Supplier B mean (66.50 g) [A1]

Supplier A eggs are heavier on average by 0.78 g.

The buyer wants the largest eggs, so should choose Supplier A.

(4 marks)

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Q12 A man has £5000.

He will put his money into a savings account at a bank.

He wants to save it for one year.

Bank A pays 2% compound interest. It adds interest every 6 months.

Bank B pays 3.15% annual interest rate.

Which bank should the man choose?
Explain your decision. Include calculations to support your decision.

Decision *(tick one box)* **Bank A** ☑ **Bank B** ☐

Show all your working

Bank A: 2% compound interest added every 6 months (2 periods in 1 year): [M1]

- After 6 months: £5000 × 1.02 = £5100.00
- After 12 months: £5100 × 1.02 = £5202.00 (or £5000 × 1.02^2 = £5202.00) [A1]

Total interest earned from Bank A = £5202 − £5000 = £202.00

Explanation and supporting calculations.

Bank B: 3.15% simple annual interest:

- After 1 year: £5000 × 1.0315 = £5157.50 (Interest = £157.50) [A1]

Comparison: £5202.00 > £5157.50 [A1]

Bank A yields £44.50 more interest than Bank B. Man should choose Bank A.

(4 marks)

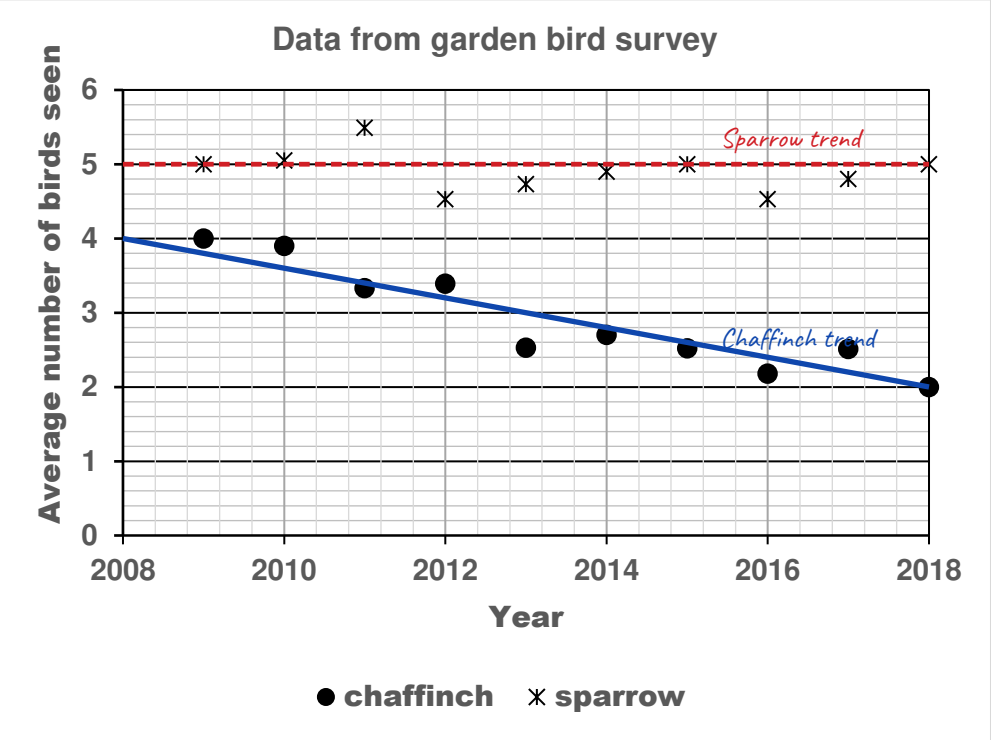
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Q13 A magazine article has this headline.

> **Garden birds in decline**
>
> **Fewer birds seen in our gardens**

An environmental organisation records information about bird populations.

The graph shows some data about two types of garden birds.



Is the magazine headline correct?

Show suitable calculations of percentage changes for the last ten years and draw trend lines on the graph to support your answer.

Explain your answer and make **two** comments.

Show all your working

Chaffinch: (2.0 − 4.0) ÷ 4.0 × 100 = −50% (50% decrease over 10 yrs) [M1]

Sparrow: (5.0 − 5.0) ÷ 5.0 × 100 = 0% (no change, stable population) [A1]

Comment 1 Chaffinch numbers declined significantly by 50% over the 10 years. [B1]

Comment 2 Sparrow numbers remained stable at ~5 birds with 0% change. [B1]

Conclusion: Headline is only partly correct (chaffinches declined, sparrows did not).

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(5 marks)

Q14 A team manager wants to pick four people for a bowling competition.

She decides to pick players with consistently good average **(median)** scoring performances.

She picks the following three people.

Player	Average score (median) over last eight matches	Range of scores over last eight matches
Archie	**105**	**26**
Baz	**101**	**37**
Cathy	**99**	**32**

She needs one more player to make up the team.
She look at the scores of two more players.

Scores in last eight matches	
Dave	**Elaine**
78	**87**
48	**98**
102	**101**
98	**84**
86	**93**
101	**79**
67	**87**
96	**97**

Make suitable calculations for Dave and Elaine.
Decide which player best fits the requirements. Explain your decision using figures.

Player picked *(tick one box)* **Dave** ☐ **Elaine** ☑

Explanation and supporting calculations

Dave scores ordered: 48, 67, 78, 86, 96, 98, 101, 102 [M1]

• Dave Median = (86 + 96) ÷ 2 = 91 • Dave Range = 102 - 48 = 54 [M1]

Elaine scores ordered: 79, 84, 87, 87, 93, 97, 98, 101

• Elaine Median = (87 + 93) ÷ 2 = 90 • Elaine Range = 101 - 79 = 22 [A1]

Comparison & Decision: Both players have similar medians (Dave 91 vs Elaine 90). [M1]

However, Elaine is far more consistent as her range of 22 is much smaller than Dave's 54.

Therefore, Elaine best fits the requirements for consistently good performance. [A1]

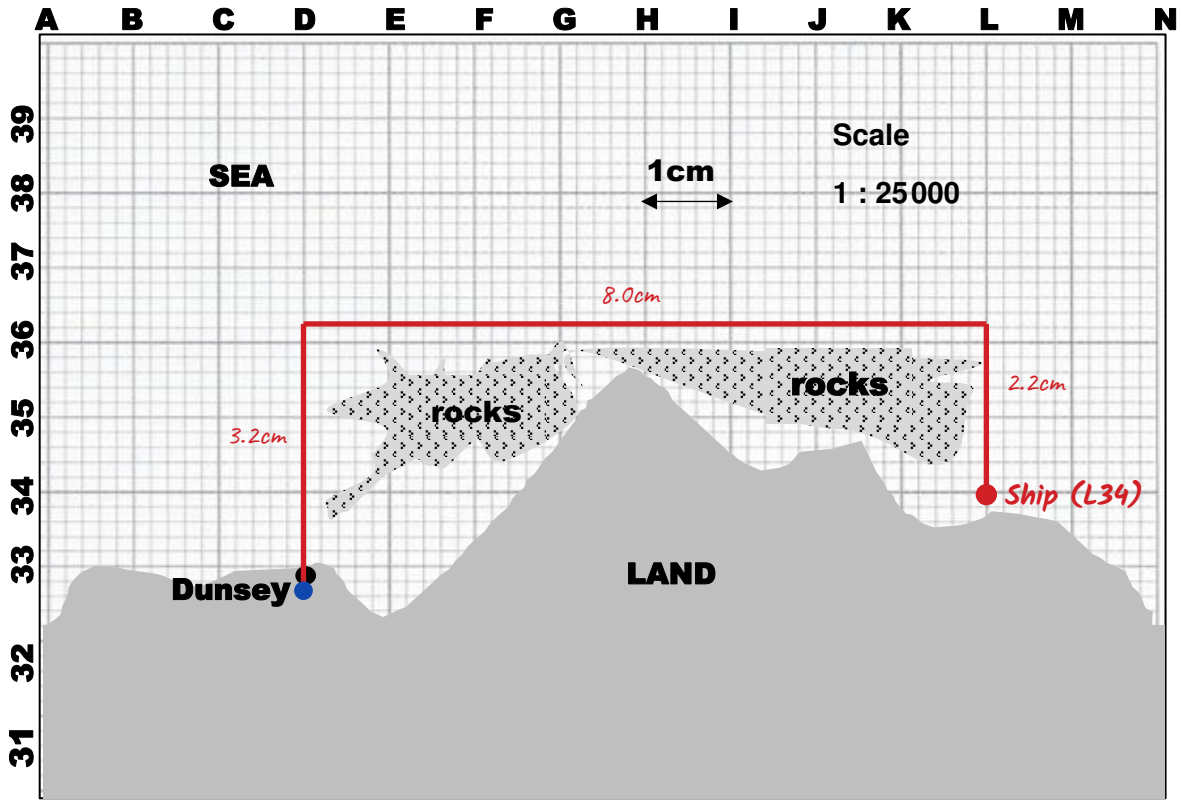
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(5 marks)

Q15 A fisherman sets off from Dunsey in a boat.

His boat will travel at an average speed of 5 kilometres per hour.

He has this map so that he can avoid the rocks.



He needs to meet up with a ship at coordinate L34 at 5pm

He needs to know at what time he must leave Dunsey.

At what time must he leave Dunsey?

Show all your working.

Step 1: Measure safe route avoiding rocks on map: [M1]

North from Dunsey: 3.2 cm · East past rocks: 8.0 cm · South to ship: 2.2 cm

Total distance on map = 3.2 + 8.0 + 2.2 = 13.4 cm (range: 12–15 cm) [M1]

Step 2: Convert to real-life distance using scale 1 : 25,000:

13.4 cm × 25,000 = 335,000 cm = 3,350 m = 3.35 km [M1]

Step 3: Calculate travel time at 5 km/h: [A1]

Time = Distance ÷ Speed = 3.35 km ÷ 5 km/h = 0.67 hours

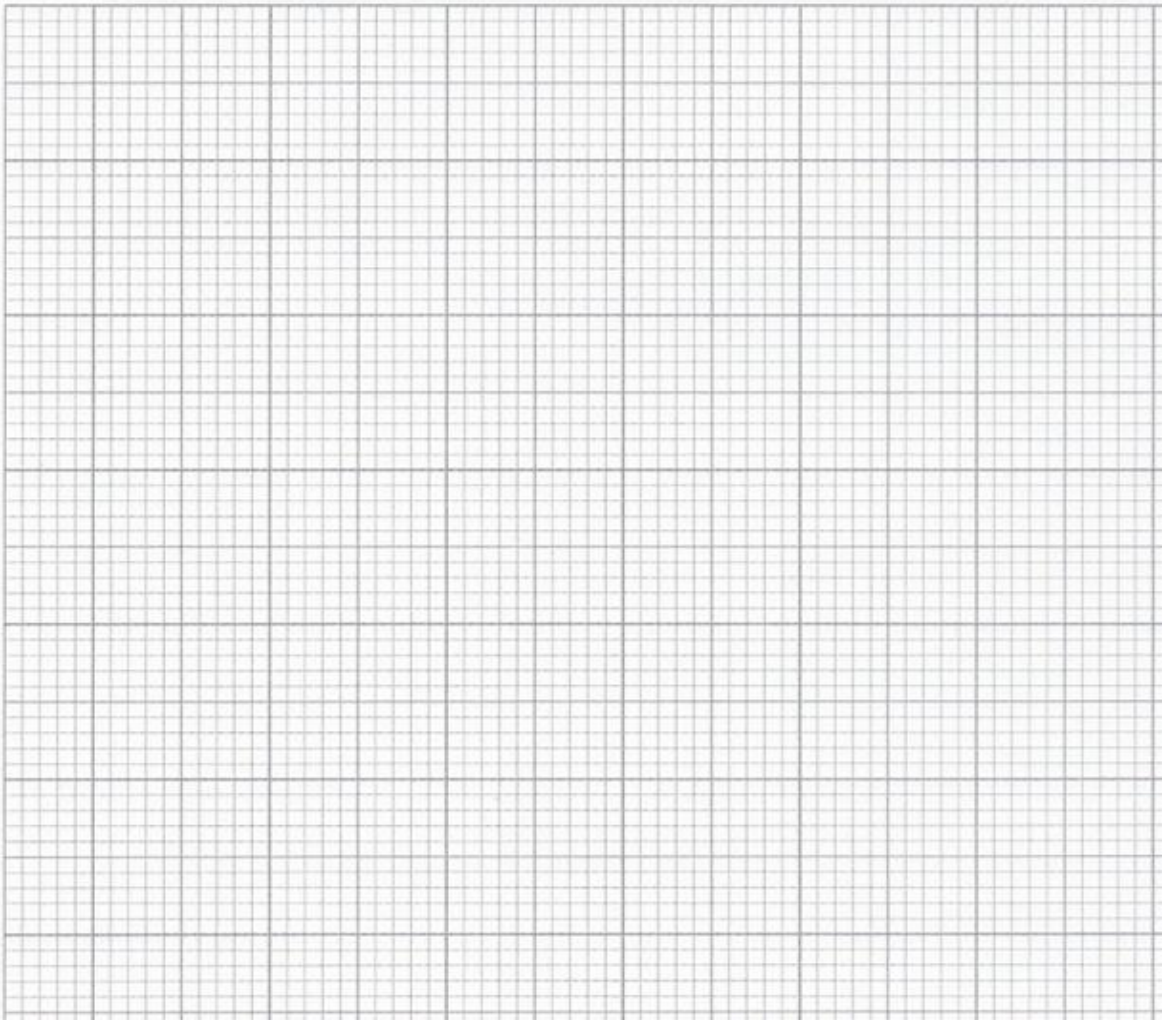
0.67 hours × 60 = 40.2 minutes ≈ 40 minutes departure before 5:00 pm [A1]

Time to leave Dunsey 4:20 pm

(6 marks)

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Spare graph paper for Question 9



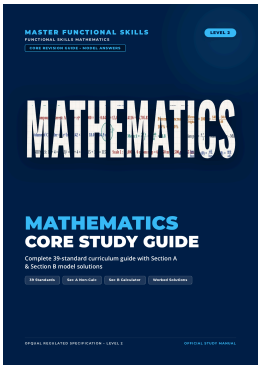
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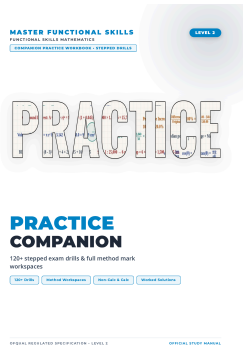


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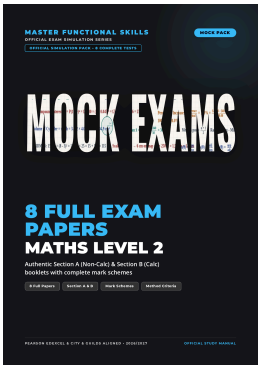


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